

Question 1:

$$f_n(x) = \begin{cases} \sin(\pi x), & \text{if } 2n \leq x < (2n+1), \\ 0, & \text{otherwise.} \end{cases}$$

a)**b)**

To test for pointwise limits we can take $x = 2.5$ as a constant value.

When $n = 1$ the interval is $[2, 3)$

$$f_1(2.5) = \sin(2.5\pi) = 1$$

When $n = 2$ the interval is $[4, 5)$

$$f_2(2.5) = 0$$

When $n = 3$ the interval is $[6, 7)$

$$f_3(2.5) = 0$$

$$\vdots$$

When $n = k$ the interval is $[2k, 2k+1)$

$$f_k(2.5) = 0$$

$$\forall x \in \mathbb{R}, \lim_{n \rightarrow \infty} = 0$$

Therefore $f_n \rightarrow f$ pointwise as $n \rightarrow \infty$

c)